

PERFORMANCE WORK STATEMENT (PWS)
CATERPILLAR 3408 DIESEL ENGINE MAJOR OVERHAUL

Part 1

General Information

1. GENERAL: This is a non-personnel service contract to provide major overhaul maintenance on a Caterpillar 3408B POWER GENERATION Diesel Engine, S/N 78Z05911. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.

1.1 Description of Services/Introduction: The contractor shall provide all personnel, equipment, supplies, facilities, shipping, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform Diesel Engine Major Overhaul as defined in this Performance Work Statement except for those items specified as government furnished property and services. The contractor shall perform to the standards in this contract.

1.2 Background: The crane's "power generation" diesel engine, Mfr. Caterpillar (CAT) 3408 has reached its recommended threshold for a Major Overhaul. This diesel was removed from an Amclyde Crane stationed at Puget Sound Naval Shipyard.

1.3 Objectives: Caterpillar OEM Certified repair facility to perform repairs/open and inspect/refurbishment/replacement on Caterpillar 3408B Power Generation Diesel Engine as described in Part 5, Specific Tasks, such that repaired equipment functions as intended. Contractor maintains close liaison and good communications with Government. Also, provide a final Condition Report of assessment.

1.4 Scope: The following work outlined in Part 5, Specific Tasks, must be accomplished by a Certified OEM engine repair facility; the repair facility shall be specifically OEM certified to repair/service "power generation" diesel generator sets, such is the model listed above. The repairs shall be performed per OEM specifications, using ONLY Genuine OEM parts. It is preferred the service personnel are ASE certified diesel mechanic(s). (T2-diesel engines, 3408B experience preferable).

1.5 Type of Contract: The government will award a Firm-fixed Price Contract.

1.6 Period of Performance: The contractor shall have 10 weeks beginning on or after the period of performance includes shipping receiving/delivery or physical pickup/delivery of the diesel to and from PSNS.

1.6.1 Quality Assurance: The government shall evaluate the contractor's performance under this contract in accordance with the Quality Assurance Surveillance Plan. This plan is primarily focused on what the Government must do to ensure that the contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rate(s).

1.7 Place of Performance: The work to be performed under this contract will be performed at the contractor's repair facility.

1.8. Pickup and Delivery. Puget Sound Naval Shipyard will be responsible for shipping of diesel to the awarded vendors location. The contractor will be responsible for shipping return of the diesel to Puget

Sound Naval Shipyard. Coordination of the shipping delivery/receive, and possible pickup/drop-off will be made with the Planning point of contact, [REDACTED], listed under 1.10, Government Point of Contact.

Puget Sound Naval Shipyard 1400 Farragut Ave Bremerton, WA 98314

1.8.1 Post Award Conference/Periodic Progress Meetings: The Contractor agrees to attend any post award conference convened by the contracting activity or contract administration office in accordance with Federal Acquisition Regulation Subpart 42.5. The Contracting Officer (KO), Contracting Officers Representative (COR), and other approved appointed Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the contracting officer will apprise the contractor of how the government views the contractor's performance and the contractor will apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the government.

1.9 PHYSICAL Security: The contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.9.1 Special Qualifications: Contractor must have the ability to test the unit fully described in Part 5, Specific Tasks. It is preferred that the service personnel are ASE certified diesel mechanic(s). (T2-diesel engines, 3408 experience preferable)

1.10 Contracting Officer Representative (COR): The COR is not authorized to modify any terms or conditions of the contract. The Contracting Officer's Representative (COR) will be identified by a separate letter. The COR oversees all technical aspects of the contract and assists with contract administration. Their responsibilities include ensuring the Contractor meets the technical requirements, conducting necessary inspections, and maintaining written and oral communication with the Contractor regarding technical matters. With the support of the designated TA/TPOC, the COR can issue written interpretations of technical specifications, including government drawings, designs, and specifications. They monitor the Contractor's performance, notify both the Contracting Officer and Contractor of any deficiencies, coordinate the availability of government-furnished property, and facilitate site access for Contractor personnel. A letter of designation sent to the COR and a copy provided to the Contractor outlines the COR's responsibilities and limitations, particularly regarding changes in cost, price, estimates, or delivery schedules.

Contracting Officer Representative: (COR):
[REDACTED]
[REDACTED]

1.11 Government Technical Points of Contact (TA/TPOC):

The TA/TPOC(s) are responsible for monitoring all technical aspects of the contract and assisting with contract administration. They are authorized to perform these duties after being appointed by the COR and having a signed TA form on file. Their responsibilities include ensuring the Contractor meets the technical requirements, helping with inspections related to contract performance, maintaining written and oral communication with the COR about technical matters, issuing written interpretations of technical requirements such as government drawings, designs, and specifications, assisting in monitoring the Contractor's performance and reporting any deficiencies to the COR, coordinating the availability of government-furnished property, and helping to facilitate site access for Contractor personnel.

The TA/POC(s) are NOT authorized to change any of the terms and conditions of this contract. If the contractor believes any Government Personnel has effectively issued a change resulting in a change in effort and price/cost of the contract, the Contractor shall immediately notify the COR or Contracting Officer.

1.12. Identification of Contractor Employees: NA

PART 2 DEFINITIONS & ACRONYMS

2. DEFINITIONS AND ACRONYMS:

2.1. DEFINITIONS

2.1.1. **CONTRACTOR.** A supplier or vendor awarded a contract to provide specific supplies or service to the government. The term used in this contract refers to the prime.

2.1.2. **CONTRACTING OFFICER (KO).** A person with authority to enter into, administer, and/or terminate contracts, and make related determinations and findings on behalf of the government. Note: The only individual who can legally bind the government.

2.1.3. **CONTRACTING OFFICER'S REPRESENTATIVE (COR).** An employee of the U.S. Government appointed by the contracting officer to administer the contract. Such appointment shall be in writing and shall state the scope of authority and limitations. This individual has authority to provide technical direction to the Contractor if that direction is within the scope of the contract, does not constitute a change, and has no funding implications. This individual does NOT have authority to change the terms and conditions of the contract.

2.1.4. **DEFECTIVE SERVICE.** A service output that does not meet the standard of performance associated with the Performance Work Statement.

2.1.5. **DELIVERABLE.** Anything that can be physically delivered but may include non-manufactured things such as meeting minutes or reports.

2.1.6. **KEY PERSONNEL.** Contractor personnel that are evaluated in a source selection process and that may be required to be used in the performance of a contract by the Key Personnel listed in the PWS. When key personnel are used as an evaluation factor in best value procurement, an offer can be rejected if it does not have a firm commitment from the person(s) that are listed in the proposal.

2.1.7. **PHYSICAL SECURITY.** Actions that prevent the loss or damage of Government property.

2.1.8. **QUALITY ASSURANCE.** The government procedures to verify that services being performed by the Contractor are performed according to acceptable standards.

2.1.9. **QUALITY ASSURANCE Surveillance Plan (QASP).** An organized written document specifying the surveillance methodology to be used for surveillance of contractor performance.

2.1.10. **QUALITY CONTROL.** All necessary measures taken by the Contractor to assure that the quality of an end product or service shall meet contract requirements.

2.1.11. **SUBCONTRACTOR.** One that enters into a contract with a prime contractor. The Government does not have privity of contract with the subcontractor.

2.1.12. **WORKDAY.** The number of hours per day the Contractor provides services in accordance with the contract.

2.1.12. WORK WEEK. Monday through Friday, unless specified otherwise.

2.2. ACRONYMS:

ACOR	Alternate Contracting Officer's Representative
AFARS	Army Federal Acquisition Regulation Supplement
AR	Army Regulation
CCE	Contracting Center of Excellence
CFR	Code of Federal Regulations
CONUS	Continental United States (excludes Alaska and Hawaii)
COR	Contracting Officer Representative
COTR	Contracting Officer's Technical Representative
COTS	Commercial-Off-the-Shelf
DA	Department of the Army
DD250	Department of Defense Form 250 (Receiving Report)
DD254	Department of Defense Contract Security Requirement List
DFARS	Defense Federal Acquisition Regulation Supplement
DMDC	Defense Manpower Data Center
DOD	Department of Defense
FAR	Federal Acquisition Regulation
GFP	Government Furnished Product
HIPAA	Health Insurance Portability and Accountability Act of 1996
KO	Contracting Officer
OCI	Organizational Conflict of Interest
OCNUS	Outside Continental United States (includes Alaska and Hawaii)
ODC	Other Direct Costs
PIPO	Phase In/Phase Out
PIEE	Procurement Integrated Enterprise Environment
POC	Point of Contact
PRS	Performance Requirements Summary
PWS	Performance Work Statement
QA	Quality Assurance
QAP	Quality Assurance Program
QASP	Quality Assurance Surveillance Plan
QC	Quality Control
QCP	Quality Control Program
TE	Technical Exhibit

2.1.1. All Definitions unless defined within the Contract are as defined within FAR Part 2.

PART 3 GOVERNMENT FURNISHED PROPERTY, EQUIPMENT, AND SERVICES

3. GOVERNMENT FURNISHED ITEMS AND SERVICES:

3.1. **Equipment:** The Government will provide submitting the Caterpillar 3408B POWER GENERATION Diesel Engine, S/N 78Z05911 into the PIEE GFP process.

3.1.1 Contractors are required to report receipt and transfer of all GFP through PIEE.

- Contractors guide on PIEE on how to obtain access.
<https://piee.eb.mil/xhtml/unauth/web/homepage/vendorCustomerSupport.xhtml>

- I. The Contractor will be Property Shipper, Contractor Property Receiver, and Contractor Property Manager Roles.
- You can find all trainings on the GFP Module at <https://dodprocurementtoolbox.com/>

PART 4 CONTRACTOR FURNISHED ITEMS AND SERVICES

4. CONTRACTOR FURNISHED ITEMS AND RESPONSIBILITIES:

4.1 General: The Contractor shall furnish all supplies, equipment, facilities and services required to perform work under this contract that are not listed under Part 3 of this PWS.

4.3. Materials: The Contractor shall furnish their own tools, test equipment, parts and ancillary equipment needed to complete the work described in Part 5, Specific Tasks.

PART 5 SPECIFIC TASKS

5. SPECIFIC TASKS:

5.1. Basic Services. The contractor shall provide services outlined for a Major Overhaul of a Cat 3408 POWER GENERATION diesel.

5.2. Tasks

The following work shall be accomplished by a Certified OEM engine repair facility and Power Systems Certified; the repair facility shall be specifically OEM certified to repair/service “power generation” diesel generator sets, such is the model listed above. The repairs shall be performed per OEM specifications, using ONLY Genuine OEM parts. It is preferred the service personnel is ASE certified diesel mechanic(s). (T2-diesel engines, 3408 experience preferable).

The *PSNS desired outcome* of the overhaul is that the engine is completely rebuilt with new parts (new/remanufactured/reconditioned) to ensure that there is a lesser probability for additional engine work or failure in the near future due to reusing “suitable” old parts. This is to avoid the extremely expensive and time-consuming work to remove and replace a crane engine

5.2.1 Disassembly and Inspection:

Disassembly and Inspection of provided Cat 3408 POWER GENERATION diesel, to include:

- A. Perform initial inspection, to include:
 - a. Disassembly of all components.
 - b. Visually inspection of all components.
- B. Submit findings to PSNS. Additionally, provide any recommendations for repair beyond the items authorized for repair in 5.2.2 to include a price quote and lead time to repair.

B.1. Do not repair additional items not identified within section 5.2.2 unless approved in writing by Contracting Officer (KO).

5.2.2 Repair Actions:

- A. Clean engine on arrival.

- B. Clean all components. Perform disassembly and inspection directions above (5.2.1)
- C. All parts replaced will be OEM industry standards.
- D. Perform the following repair actions: Rebuild the engine: replace all the listed parts and any additional parts needed that are not listed here to achieve the ***PSNS desired outcome***:

Tasks for the Major Overhaul:

1. CYLINDER HEADS (RECONDITION/REMAN)
2. TURBO CARTRIDGES (RRECONDITION/REMAN)
3. FUEL INJECTORS (REMAN)
4. REPLACE INJECTION PUMP (REMAN)
5. JW WATER PUMP (REMAN)
6. REPLACE JACKET WATER TEMP REGULATORS (NEW)
7. AUX WATER PUMP (REMAN)
8. WATER TEMP REGULATORS (THERMOSTAT) (NEW)
9. AFTERCOOLER CORE (CLEAN/TEST)
10. OIL SAMPLE (TEST/REPORT)
11. OIL FILTERS (NEW)
12. FUEL FILTERS (NEW)
13. ALL REQUIRED GASKETS, SEALS, O-RINGS (NEW)
14. PAINT BARE COMPONENTS: Items 1-13 PER ABOVE PARTS LIST
15. CYLINDER PACKS (RODS/LINERS/PISTONS/RINGS)
16. ENGINE OIL PUMP (REMAN)
17. AFTERCOOLER CORE (RECONDITIONED/REMAN)
18. ENGINE OIL COOLER CORE (REMAN)
19. FUEL TRANSFER PUMP (REMAN)
20. CRANKCASE BREATHERS AND CLAMPS (NEW)
21. CRANKSHAFT MAIN BEARINGS AND THRUST PLATES (NEW)
22. INSPECT AND REPLACE CRANK BALANCER IF REQUIRED
23. CONNECTING ROD BEARINGS (NEW)
24. ROCKER ARMS ASSEMBLIES - INTAKE/EXHAUST
25. AIR CLEANER ELEMENT (NEW)
26. CAMSHAFTS AND LIFTERS (REMAN)
27. CAMSHAFT BUSHINGS/BEARINGS
28. REPLACE CAMSHAFT BALANCER ASSEMBLY
29. FRONT GEAR TRAIN GEARS AND BUSHINGS
30. REAR GEAR TRAIN GEARS AND BUSHINGS
31. FRONT AND REAR MAIN CRANKSHAFT SEALS
32. VIBRATION DAMPER
33. PISTON COOLING JETS
34. COOLANT HOSES
35. INSPECT CRANKSHAFT UPON TEAR DOWN
36. INSTALL ADDITIONAL PARTS LISTED:

- 1ea - 4P-3864 COVER
- 1ea - 4N0362 COVER A
- 1ea - 7C-0307 GASKET
- 1ea - 4N1156 GASKET
- 1ea - 6N1410 ELBOW1
- 1ea - 6N1411 ELBOW1
- 1ea - 6N1656 GOV DRIVE G
- 1ea - 4N7651 PIPE
- 1ea - 1W9070 DIODE A

- 1ea - 2W7553 MANIFOLD
- 1ea - 4N6145 SUPPORT
- 1ea - 1052508 PUMP AS
- 1ea - 1736996 SOLENOID AS
- 1ea - 1342252 SENSOR GP EL
- 1ea - 4499582 KIT-GASKET
- 1ea - 6N9076 TUBE "A" OIL
- 2ea - 1078106 MANIFOLD
- 2ea - 1078112 MANIFOLD
- 1ea - 155-2260 PLUG KIT-CON
- 1ea - 108-3190 SWITCH
- 3ea - 8T-8730 CONNECTOR
- 1ea - 7E3903 TUBE
- 1ea - 7E3904 TUBE
- 1ea - 7E3902 TUBE
- 1ea - 7E3901 TUBE

- E. All seals and gaskets will be replaced with new and all hardware torqued to CAT spec.
- F. Valves and fuel system timed and adjusted to CAT guidelines.
- G. Upon approval by the Contracting Officer via a contract modification, perform recommended additional repair items identified under 5.2.1-B.
- H. After completion of repairs, perform the minimum testing listed below:
 - a. Test run engine to include initial break-in
 - b. Perform engine dyno at all engine RPMs and record HP and Torque for customer records
 - c. Test and report on the condition of Safety devices.
- I. Once successful testing for normal operation, return the completed engine to Puget Sound Naval Shipyard.

TECHNICAL EXHIBIT 1

Performance Requirements Summary

The contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success.

Performance Objective (The Service required—usually a shall statement)	Standard	Performance Threshold (This is the maximum error rate. It could possibly be “Zero deviation from standard”)	Method of Surveillance
PRS # 1. The contractor shall provide a complete Major Overhaul	The contractor provides 100% compliance	Zero deviation from the <i>Caterpillar Engine Manual SENR7377</i>	<i>Contractor shall provide assessment report to COR</i>
PRS # 2 Contractor shall maintain close liaison and good communication with government.	Any problems or questions encountered by either party are addressed within 24 hours (1 business day)	N/A	<i>Provide COR with written report of finding/discrepancies</i>
PRS # 3 Contractor shall provide complete final Condition Report assessment.	The contractor provides 100% compliance	N/A	<i>Provide COR with the final Condition Report</i>

TECHNICAL EXHIBIT 2

DELIVERABLES SCHEDULE

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
Condition found report of crank balancer, rocker arms, camshaft and lifters, crankshaft, to determine if replacement is recommended through visual inspection	Report findings and replacement recommendations within 24 hours	1	PDF	COR
Unexpected Findings Report will be submitted for items not listed in Tasks or that do not meet CAT guideline for reuse upon inspection.	Document, and email recommended corrective actions	1	PDF	COR
Test and Report Condition of Safety Devices	Two weeks after acceptance	1	PDF	COR
Fuel pump Test report	Two weeks after Performance Test	1	PDF	COR
Final Engine Dyno Performance Test Results	Two weeks after Performance Test	1	PDF	COR
Report on components replaced	Two weeks after completion of service	1	PDF	COR
Any Applicable Engine Settings	Two weeks after completion of service	1	PDF	COR